

Convert metric lengths

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Convert 4.375 km to metres. Copy or complete the first step.

2. Convert 2.65 kg to grams and 3,750 ml to litres.

3. Write 845 mm in metres.

4. Miro says: Miro multiplies whenever converting, regardless of the direction between units. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Convert 4.375 km to metres. Copy or complete the first step.
Answer 4,375 m.
2. Convert 2.65 kg to grams and 3,750 ml to litres.
Answer 2,650 g and 3.75 litres.
3. Write 845 mm in metres.
Answer 0.845 m.
4. Miro says: Miro multiplies whenever converting, regardless of the direction between units. Is this correct? Explain.
Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.

Convert metric lengths

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. Convert 4.375 km to metres.

6. Check this result using an inverse, estimate or known fact: Write 845 mm in metres.

2. Convert 2.65 kg to grams and 3,750 ml to litres.

7. Prove or explain your reasoning: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

3. Write 845 mm in metres.

8. Identify and correct this misconception: Miro multiplies whenever converting, regardless of the direction between units.

4. Solve this independently and show every stage: Convert 4.375 km to metres.

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Convert 2.65 kg to grams and 3,750 ml to litres.

10. Write and solve a similar question to: Write 845 mm in metres.

Teacher answers and checking prompts

1. Convert 4.375 km to metres.
Answer 4,375 m.
2. Convert 2.65 kg to grams and 3,750 ml to litres.
Answer 2,650 g and 3.75 litres.
3. Write 845 mm in metres.
Answer 0.845 m.
4. Solve this independently and show every stage: Convert 4.375 km to metres.
Answer 4,375 m.
5. Use a second method or representation for: Convert 2.65 kg to grams and 3,750 ml to litres.
Answer Expected result: 2,650 g and 3.75 litres. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: Write 845 mm in metres.
Answer Expected result: 0.845 m. A valid check must be shown.
7. Prove or explain your reasoning: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.
Answer About 48 km. The explanation must show why the method works.
8. Identify and correct this misconception: Miro multiplies whenever converting, regardless of the direction between units.
Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: Write 845 mm in metres.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Convert metric lengths

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.

6. Create a new example that tests the same idea as: Write 845 mm in metres.

2. Prove the answer to this question, rather than only calculating it: Convert 4.375 km to metres.

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Convert 2.65 kg to grams and 3,750 ml to litres.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 0.845 m.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro multiplies whenever converting, regardless of the direction between units.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Use 5 miles is about 8 km. Estimate 30 miles in kilometres.
Answer About 48 km. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: Convert 4.375 km to metres.
Answer Expected result: 4,375 m. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Convert 2.65 kg to grams and 3,750 ml to litres.
Answer Expected result: 2,650 g and 3.75 litres. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 0.845 m.
Answer One valid reconstruction is: Write 845 mm in metres. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro multiplies whenever converting, regardless of the direction between units.
Answer Use the size of the target unit to decide the direction, then check whether the numerical value should grow or shrink.
6. Create a new example that tests the same idea as: Write 845 mm in metres.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.